TUTORIAL – 1

1. Music is stored in an audio CD that has 2 channels at a sampling rate of 44.1 kHz and each sample is encoded using 16 bits. If the total memory space consumed by the music is 700 MB, determine the duration of the music stored. (Ans: 66.1minutes)

2. Convert the number $(1110.101)_2$ to decimal. (Ans: $(14.625)_{10}$)

3. The frequency spectrum of a speech signal is shown in Fig. 1-1. What is the minimum sampling frequency required to be used for sampling the speech to ensure that *aliasing* does not occur? (Ans: 9.6 kHz)

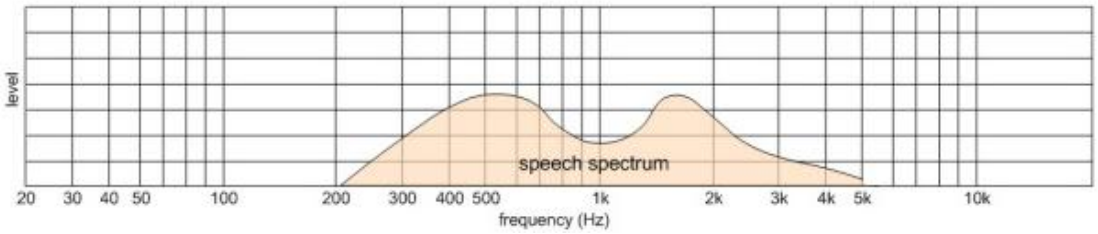

Figure 1-1

4. Refer to Fig. 1-2.

   a) What is the frequency of the original analog signal?
   (Ans: 3 KHz)
   b) What is the theoretical minimum sampling frequency required to avoid aliasing?
   (Ans: 6 KHz)
   c) The signal is sampled as shown in the figure where the small circles represent the sampled data. What is the sampling frequency used? Does the sampling follow Sampling Theorem?
   (Ans: Sampling frequency = 6 kHz. It is in line with the sampling theorem)
   d) Comment on the frequency and amplitude of the signal reconstructed from the sampled signal (red wave) and compare it with that of the original signal.
   (Ans: Frequency =3KHz, amplitude and phase are different. Aliasing has occurred)
   e) If aliasing has occurred, state the possible reason.
   (Ans: Sampling is not done in phase with the analog signal)
   f) Give your recommendation to avoid aliasing in this scenario.
   (Ans: Use a slightly larger sampling frequency, i.e., Fs > fsignal (example: 8 kHz)

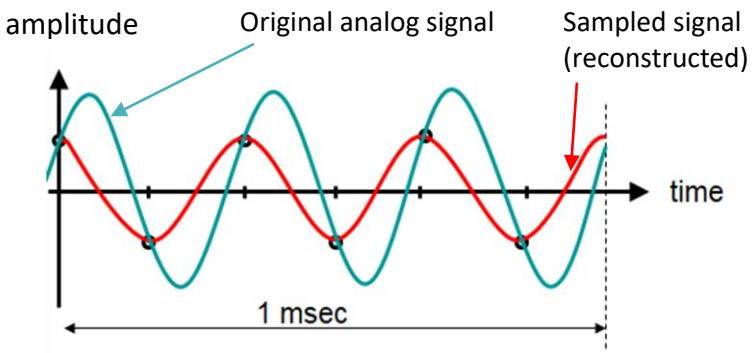

Figure 1-2

5. Refer to Fig. 1-2. Assume that the sampling frequency used is 20 kHz.

   (a) Will aliasing occur? (Ans: No)
   (b) What are the overheads incurred in using a larger sampling frequency of 20 kHz compared to 8 kHz?
   (Ans: More samples to process, low speed, high power consumption)

6. An 8-bit Analog-to-Digital Converter (ADC) is used to digitize an analog signal of Full-Scale span +6 volts to -6 volts.

   (a) Determine the worst-case Quantization error of the ADC. (Ans: 23.4375 mV)
   (b) What percentage of the Full-Scale span will be worst-case Quantization error?
   (Ans: 0.1953%)

7. Refer to Question No. 6. For a specific application, it is required to ensure that the percentage of the worst-case Quantization error is NOT more than 0.1%. What is the minimum word length (number of bits) of the ADC you would choose to meet the design specification? (Ans: 9 bits)

8. Convert the following numbers with the indicated bases to decimal:

   (a) $(4310)_8$ (Ans: $(2248)_{10}$)
   (b) $(1ABC)_{16}$ (Ans: $(6844)_{10}$)

9. How many bits are needed to represent $(1ABC)_{16}$ in binary? (Ans: 13 bits)

10. What is the largest Hexadecimal number using 4 digits? Find its decimal equivalent.
    (Ans: (FFFF)16 = (65535)10)

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